

CHAPTER 3

Constructing Tests and Evaluating Weakness

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One-Shot Weakness Detection

Where does a fixed model perform poorly?

1 FIXED MODEL

Keep the predictor fixed

We are not retraining the model.

We are changing where validation looks for error.

2 OBSERVATION-LEVEL ERROR

Compute loss

For each observation:

$$\text{Brier} = (y - p)^2$$

This gives a row-level error signal.

3 WEAK REGION

Find concentrated loss

Partition the population and compare loss inside each group.

A weak region has materially higher loss.

4 TEACHING POINT

A model can look acceptable overall while hiding a pocket of concentrated error.

The one-shot assessment makes “weakness” visible before we discuss adaptive search.

Detailed Example: XGBoost Error Model + Clustering

Use row-level Brier loss to learn a geometry where errors concentrate.

1 PIPELINE



2 WHAT XGBOOST CONTRIBUTES

Error-aware representation

Fit XGBoost to squared residuals / Brier loss.

Customers are embedded by their per-tree leaf outputs.

Clustering in error geometry

Cluster customers using similarity in fitted error patterns, not just raw customer features.

Loss comparison

For each cluster, compute the fixed model's Brier loss and compare it with the rest.

Generalizing the Example: Inner Array × Outer Array

1 INNER ARRAY

What we vary

Candidate clusters, medoid sets,
region rules or task presentations.

This is the search space.

2 OUTER ARRAY

Where we test

Common samples, populations,
vintages, stresses or source-
access settings.

This is the evaluation basis.

3 EVALUATOR OUTPUT

What we measure

Scores, support, uncertainty,
evaluation status and traceable
records.

These drive comparison.

From One-Shot to Adaptive Discovery

The first result is useful; the next question makes it a loop.

1 ONE-SHOT RESULT

Initial weakness table

The initial clustering gives measured regions and coverage.

It may already reveal a material concern.

2 NEXT QUESTIONS

What should we test next?

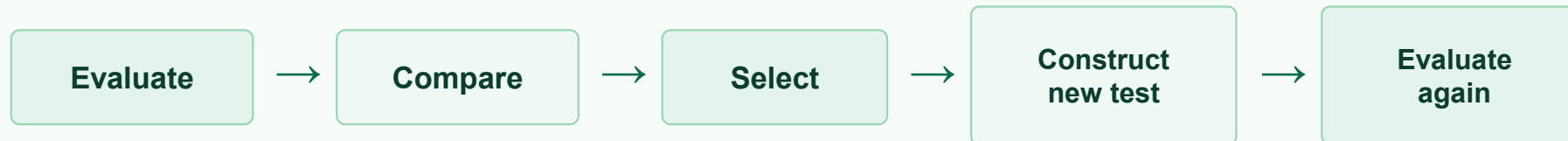
Is the weakness narrower?
Does it persist in another vintage?
Is there a nearby region with larger loss?

3 ADAPTIVE LOOP

Measurement drives search

Use the evidence to refine candidates, allocate budget and build the next executable test.

4 LOOP LOGIC



One-shot becomes discovery when results determine the next test.

Why Pareto? Preserve Different Weaknesses

A weak region can be “bad” in more than one way.

1 TWO OBJECTIVES IN WEAKNESS DETECTION

Average loss

How poorly the model performs on average in the candidate region.

Worst-case / tail loss

How bad performance gets in the hardest part of the region.

Coverage / support

How much of the population the weakness affects and whether it is measurable.

2 ILLUSTRATIVE CANDIDATES

Region	Avg. Brier	Worst case	Interpretation
A	0.055	0.11	Broad average weakness
B	0.045	0.18	Severe tail weakness
C	0.052	0.16	Tradeoff worth retaining
D	0.035	0.09	Dominated by stronger candidates

3 KEY PRINCIPLE

Pareto keeps distinct failure modes visible instead of hiding them inside one arbitrary weighted score.